

|    |   |                                                                                                                                                                                                                                                                                                                                                        |
|----|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12 | D | <p>For a satellite at Earth's surface,</p> $mg = mr\omega^2 = mr\left(\frac{2\pi}{T_E}\right)^2$ $(T_E)^2 = \frac{4\pi^2 R}{g}$ <p>For a satellite at Moon's surface</p> $(T_M)^2 = \frac{4\pi^2 \left(\frac{R}{4}\right)}{\frac{g}{6}} = \frac{6}{4} \left(\frac{4\pi^2 R}{g}\right) = \frac{3}{2} (T_E)^2$ $T_M = \sqrt{\left(\frac{3}{2}\right)} T$ |
| 13 | B | <p>RV = <math>\frac{1}{2}kx</math> RT</p>                                                                                                                                                                                                                                                                                                              |